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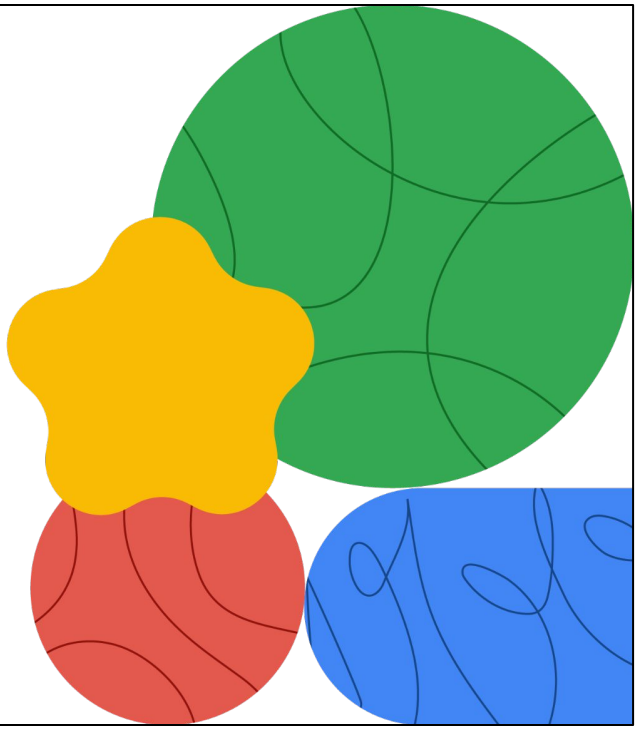
Manage Scalable Workloads in GKE



Say: Welcome to the “Manage Scalable Workloads in GKE” course.

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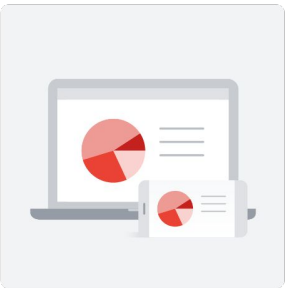
Course format



Presentations

Quizzes

Hands-on labs

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Through a combination of presentations, quizzes, and hands-on labs, you'll get comfortable with GKE architecture and practice building your own containerized applications.

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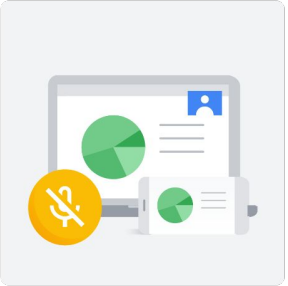
Etiquette



Mute microphone

No recording

Ask questions

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Before we get too far, let's take a moment to acknowledge some housekeeping.

1. If you're not speaking, please mute your microphone. I'd love to see your face, however, so please take a moment to turn your camera on.
2. No recording is allowed.
3. Please feel free to ask questions!

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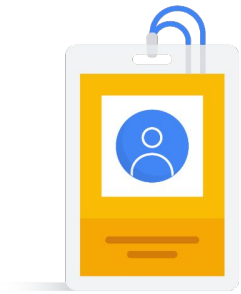
Learner profile



This course was designed for:

- Developers
- DevOps Engineers
- Cloud Architects

With intermediate knowledge of GKE.

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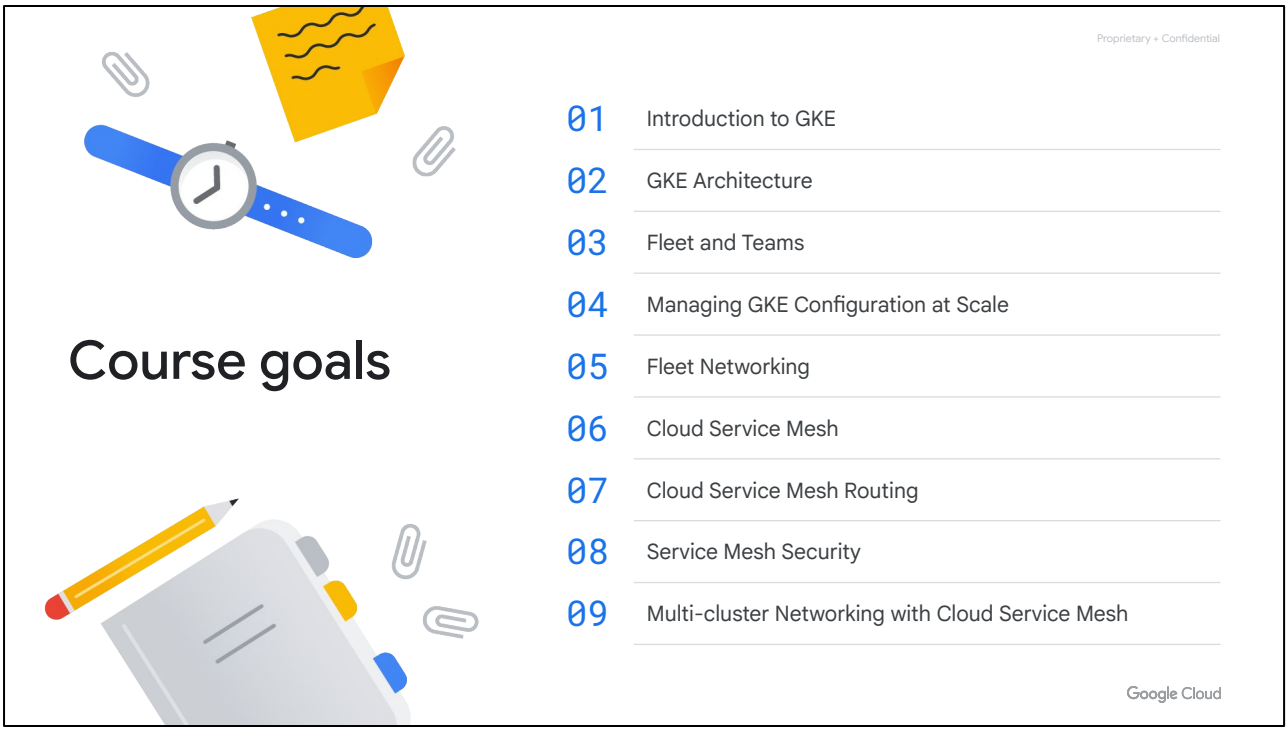
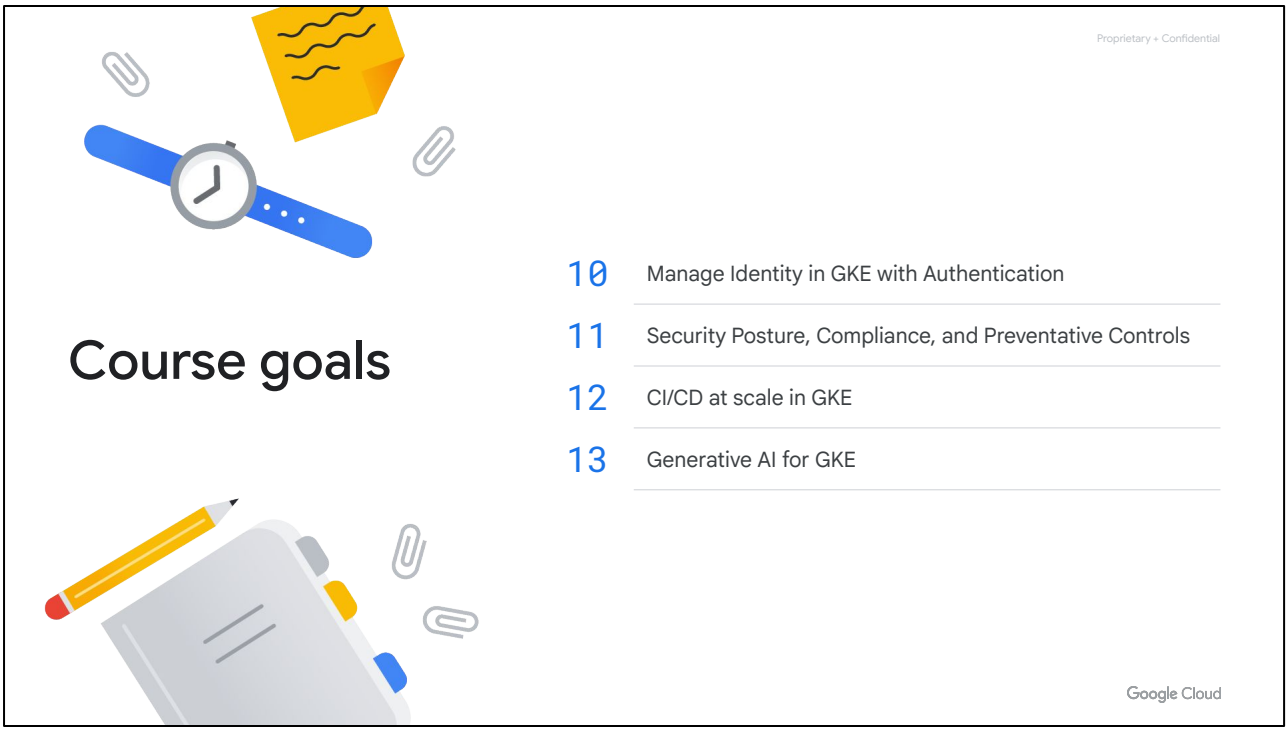
This course was designed for developers, DevOps engineers, and cloud architects with intermediate knowledge of GKE.



The learning material is spread across three days. Let's explore the overall goals.

- You'll begin with an introduction to GKE.
- From there, you'll learn about GKE architecture, followed by an introduction to fleets and teams.
- You'll then learn how to manage GKE configuration at scale and explore fleet networking.
- Then you'll transition and explore Cloud Service Mesh, including routing and security.
- You'll also explore multi-cluster networking with Cloud Service Mesh.



- From there, you'll transition to identity management with authentication, and security posture, compliance, and preventative controls.
- And finally, you'll learn how to use CI/CD at scale and leverage generative AI capabilities in GKE.

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Day 1 agenda

- Introduction to GKE at Scale
- Multi-cluster GKE Architecture
- Fleet and Teams
- Lunch
- Managing GKE Configuration at Scale
- Fleet Networking

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- ease adjust the times depending on when the class starts.

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Day 2 agenda

- Cloud Service Mesh
- Cloud Service Mesh Routing
- Lunch
- Cloud Service Mesh Security
- Multi-cluster Networking with Cloud Service Mesh

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Day 3 agenda

- Manage Identity in GKE with Authentication
- Security Posture and Preventative Controls
- Lunch
- CI/CD at scale in GKE
- GKE and AI

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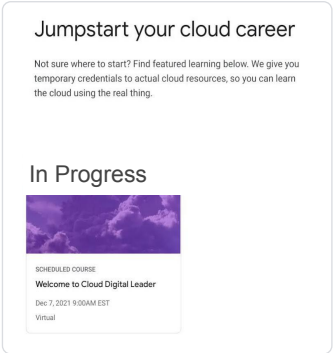
Open Qwiklabs

01 **Open an incognito window** (or private/anonymous window).

02 Go to the Qwiklabs URL cloudskillsboost.google

03 Sign in with existing account, or join with new account, using the **email you used to register** for the course.

04 Launch the course from **In Progress** section of your home page.



Access issues

If you encounter any problems while setting up your Qwiklabs account, follow the troubleshooting steps here: http://traininghelp.cloudlearning.io

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Go ahead and open Qwiklabs:

1. **Open an incognito window** (or private/anonymous window). Use of an incognito browser window reduces the risk that you will accidentally do the labs using your own Google Cloud account instead of Qwiklabs.
2. **Go** to the Qwiklabs URL your instructor provides.
3. **Sign** in with an existing account or **Join** with a new account (with email you used to register for the course).
4. Launch the course from the section of your **In Progress** home page.

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Facilitator reference links:

- Qwiklabs URL: https://googlecloud.qwiklabs.com
- Troubleshooting: http://traininghelp.cloudlearning.io

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View course materials: workbook and slides

Labs | Lecture Notes

01

02

03

04

You can download these as PDF files

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Within the course, you can also access the student slides. You can download these as PDF files.

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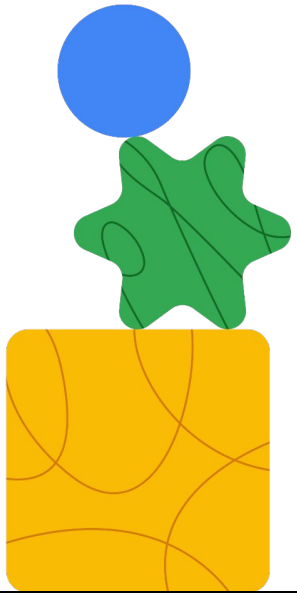
Icebreaker

Discussion

5 mins | Class

01 Take **1 minute** to think of a product or technology that no longer exists.

02 Our goal is to uncover the most interesting or unusual product or technology.



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I'd like to start with an icebreaker to know each other better.

1. Take 1 minute to think of a product or technology that no longer exists.
2. When you're ready, unmute and chat amongst yourselves about your ideas.
3. Our goal is to uncover the most interesting or unusual product or technology.